MILITARY SPECIFICATION

HEAD ASSEMBLY: 11727486

(PERISCOPE M32E1 and M35E1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a modular unit of the Tank Gunner's M32E1 and M35E1 Periscopes, herein referred to as Head Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army, Frankford Arsenal

F5549108	Telescope, Collimation (3X)
F8247391	Test Fixture for Periscope
F8565556	Pressure Tester, Two-Station
F10547987	Set-Up Instructions for Calibration Standards, Periscope M32
D10548766	Calibration Standard, 26° Elevation,
F10548767	Support Assembly Collimator
F11727486	Head Assembly
D11729723	Gage, Minimum Interchangeability
D11732101	Gage, Maximum Interchangeability
D11732107	Gage, Maximum Minimum Interchangeability
D11732108	Gage, Minimum Interchangeability
F11732155	Gage, Minimum Interchangeability
F11732156	Gage, Maximum Interchangeability
F11738140	Final Acceptance Inspection Equipment

PACKAGING DATA SHEET

PS11727486	Packaging of Head Assembly 11727486

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The head assembly shall be manufactured in accordance with Drawing F11727486 and drawings pertaining thereto.

3.2 Interchangeability.

a. Head Assembly with modules (Elbow Assembly, Passive and Body Assembly, Daylight).- The head assembly shall be capable of the insertion and removal of the passive and daylight modules without restriction (gages are specified in lieu of modules in 4.6.12.1).

b. Head Assembly with mount.- The head assembly shall be capable of insertion into the Mount Periscope: M118E1 (gages are specified in lieu of mount in 4.6.12.2) after insertion, the head assembly shall be capable of being firmly secured and subsequently removed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The head assembly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (plus 60° to 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80° to plus 160°F.

3.4.2 Operating temperatures.- The head assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at ambient temperatures of minus 40° and plus 150°F.

3.5 Operability, elevation input coupling.- The running torque required to rotate the input coupling through maximum elevation and depression shall not exceed 6 pound-inches at standard ambient temperature (plus 60° to 90°F). When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 10 pound-inches.

3.6 Orientation.- The requirements of 3.7, 3.10 and 3.11 shall be met with the head assembly positioned as follows:

a. Datum Plane "Y" on a horizontal plane
b. Side "L" of the input coupling positioned 22 degrees 30 minutes with respect to a plane defined by the four mounting pads as measured in a vertical plane.

NOTE: The following references apply throughout this specification:

Datum Plane "W" - See Drawing F11727077

3.7 Collimation.- The requirements of 3.7.1 to 3.7.8 inclusive, shall be met with:

a. An incident line of sight which is originally perpendicular to a plane defined by the three mounting pads.

b. The input coupling positioned in accordance with 3.6.

3.7.1 Pre-vibration/Pre-shock, daylight viewing area.- The incident line of sight shall emerge perpendicular to Datum Plane "Y" within 0.20 mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

3.7.2 Pre-vibration/Pre-shock, passive viewing area.- The incident line of sight shall emerge perpendicular to Datum Plane "Y" within 0.20 mil.

3.7.3 Unity power area.- The reflected line of sight shall be depressed 26 degrees below Datum Plane "Y" within 15 minutes, and be perpendicular to Datum Plane "W" within 15 minutes.

3.7.4 Excursion range.- Rotation of the input coupling shall cause the line of sight to deflect a minimum of 22.0 degrees above to 18.0 degrees below the horizontal plane.

3.7.4.1 Coupling travel.- Rotation of the input coupling shall be limited by contact with the coupling stop pin only.

NOTE: At no time shall the input coupling be allowed to strike hard against the travel stops.

3.7.5 Plumb travel.- Starting with the line of sight in coincidence with a vertical target line at the horizontal position, the line of sight shall track the target line within 0.25 mil when elevated and depressed through the range specified in 3.7.4.

3.7.6 Elevation accuracy.- This requirement shall be met with backlash removed and the angle measured in a vertical plane. The angle formed by the line of sight and side "L" of the input coupling shall remain constant within 0.20 mil at any elevation setting from 18 degrees depression to 22 degrees elevation.

3.7.7 Backlash.- Backlash between the line of sight and the input coupling shall not exceed 0.05 mil.

3.7.8 Image tilt.- This requirement shall be met when viewing through either the daylight or passive exit aperture. The image of a target line which is originally perpendicular to Datum Plane "Y" shall emerge perpendicular to Datum Plane "W" within 30 minutes of arc.

3.8 Resolution.

3.8.1 Daylight and passive viewing areas.- Resolution shall be 10 seconds of arc or less when viewing through either the daylight or passive exit apertures with a telescope having a 1.6 inch entrance pupil. Resolution consists of detectable line structure and proper line count in all four meridians at one focus setting.

3.8.2 Unity power system.- Resolution shall be 45 seconds of arc or less when viewing through the unity-power exit aperture with an observation telescope of at least 1/2 inch entrance pupil.

3.9 Vibration/Shock.

3.9.1 Vibration "A".- Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. The vibration shall be sinusoidal as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .65 G (peak).

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .65 to 1.5 G's (peak).

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's (peak).

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's (peak).

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .65 G (peak).

f. At 5 Hz, the acceleration shall be .65 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. The head assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive.

3.9.2 Vibration "B".- Vibration shall be applied along each of three mutually, perpendicular axes of the head assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G (peak).

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G (peak).

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G (peak).

d. At 5 Hz, the acceleration shall be 1.0 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minute ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes). The body assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive.

3.9.3 Shock.- The head assembly shall be subjected to 24 cumulative shocks consisting of three each at 100 G's, then 40 G's, applied in each direction along each of the two mutually perpendicular horizontal axes (all four sides). The head assembly shall also be subjected to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

six cumulative shocks consisting of three each at 90 G's, then 35 G's applied in one direction along the vertical axis (normal operating position of head assembly only). The shock pulse shall have a half sine wave. For 100 G's ± 10% and 90 G's ± 10% the shock pulse shall have a time duration of 3 ± .2 milli-seconds; for 40 G's ± 10% and 35 G's ± 10% the shock pulse shall have a time duration of 18 ± 3 milliseconds. The head assembly shall show no evidence of physical failure subsequent to shock. Following the shock test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive.

3.10 Post-vibration/Post-Shock, daylight viewing area.- The incident line of sight shall remain within 0.10 mil of the collimation reading established in 3.7.1.

3.11 Post-vibration/Post-shock, passive viewing area.- The incident line of sight shall remain within 0.10 mil of the collimation reading established in 3.7.2.

3.12 Sealing.- The head assembly shall show no evidence of leakage in excess of 0.2 pounds per square inch (psi) when subjected to an internal pressure of 5. ± .010 psi for a minimum of 2 hours. The interior atmosphere of the accepted head assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig).

3.13 Cleanliness.

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end (including the unity power system).

b. Dirt.- There shall be no more than three particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three head assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Head Assemblies.- Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232, and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.3, the tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.3.2.3 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Interchangeability, Head Assembly with modules (Elbow Assembly, Passive and Body Assembly, Daylight)	3.2a	4.6.12.1
2.	Interchangeability, Head Assembly with mount	3.2b	4.6.12.2
3.	Resolution, daylight and passive viewing areas	3.8.1	4.6.5
4.	Resolution, unity power system	3.8.2	4.6.6

NOTE: The tests in 4.3.2.3 shall be conducted at a temperature between +60° and +90°F.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
101.	Operability, elevation input coupling	3.5	4.6.3
102.	Collimation, pre-vibration daylight viewing area	3.7.1	4.6.4.1
103.	Collimation, pre-vibration, passive viewing area	3.7.2	4.6.4.2
104.	Collimation, unity power area	3.7.3	4.6.4.3
105.	Excursion range	3.7.4	4.6.4.5
106.	Coupling travel	3.7.4.1	4.6.4.6
107.	Plumb travel	3.7.5	4.6.4.7
108.	Elevation accuracy	3.7.6	4.6.4.8
109.	Backlash	3.7.7	4.6.4.4
110.	Image tilt	3.7.8	4.6.4.9
111.	Vibration "B"	3.9.2	4.6.7.2
112.	Collimation, post-vibration daylight viewing area	3.10	4.6.8
113.	Collimation, post-vibration, passive viewing area	3.11	4.6.9
114.	Sealing	3.12	4.6.10
115.	Cleanliness	3.13	4.6.11
MINOR:	NONE		

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One head assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing - visual
302.	General Specification	3.3	MIL-F-13926 - visual

4.4.2 Environmental.- Three head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in 4.3.2.3 and in Table I.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° and +160°F)	3.4.1	4.6.1
304.	Operating temperatures (40° and +150°F)	3.4.2	4.6.2
305.	Operability, elevation input coupling	3.4.2, 3.5	4.6.2, 4.6.3

NOTE: The test for characteristic 305 shall be conducted at -40° and +150°F.

4.4.3 Verification, follow-up.- Three head assemblies shall be randomly selected as samples from production lots at 25 percent, 50 percent and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table IV.

TABLE IV - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306.	Collimation, pre-vibration, daylight viewing area	3.7.1	4.6.4.1
307.	Collimation, pre-vibration, passive viewing area	3.7.2	4.6.4.2
308.	Vibration "A"	3.9.1	4.6.7.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

309.	Collimation, post-vibration, daylight viewing area	3.10	4.6.8
310.	Collimation, post-vibration, passive viewing area	3.11	4.6.9
311.	Collimation, pre-shock, daylight viewing area	3.7.1	4.6.4.1
312.	Collimation, pre-shock, passive viewing area	3.7.2	4.6.4.2
313.	Shock	3.9.3	4.6.7.3
314.	Collimation, post-shock, daylight viewing area	3.10	4.6.8
315.	Collimation, post-shock, passive viewing area	3.11	4.6.9

4.4.4 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Special inspection equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.5.2.3.1 Sealing.- The pressure testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance.- Special test equipment to determine performance of the periscope shall conform to the requirements of Drawing F11738140 (Final Acceptance Inspection Equipment) and all drawings pertaining thereto.

4.6 Test methods and procedures.

NOTE: At no time during the following tests shall the input coupling be allowed to strike hard against the travel stops.

4.6.1 Storage temperature.- This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature, (+60° to +90°F), and the head assembly removed from the test chamber. The head assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.7. The head assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures.- This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of Mil-F-13926 and the conditions of 3.4.2. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE in 4.6.1) to -40°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F, the head assembly shall be subjected to the test in 4.6.3 to determine compliance with the applicable requirements of 3.5 for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see NOTE in 4.6.1) to +150°F. The head assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the head assembly shall again be subjected to the test

Source: https://assist.dla.mil - Downloaded: 2022-07-30 13:37:37
Check the source to verify that this is the current version before use.

in 4.6.3 to determine compliance with the applicable requirements of 3.5 for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.1) to room ambient temperature (+60° to +90°F) and the head assembly removed from the test chamber. Head assemblies which have passed the above tests shall be subjected to the test of 4.6.4.

4.6.3 Operability, elevation input coupling.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5 shall be utilized for this test. An appropriate adapter to accommodate the coupling shall be used with the torque measuring equipment. The running torque required to rotate the coupling through its full range of travel in each direction, shall conform to the applicable requirements of 3.5 for the temperatures indicated.

4.6.4 Performance.- To determine compliance with 3.6, the following tests shall be performed with the head assembly positioned on the special test equipment specified in 4.5.2.3.2. Set-up of the test equipment and installation of the head assembly under test shall be performed in accordance with the set-up and calibration instructions contained on Drawing F8247391. The Collimator Projector shall be focused at 1200 meters. Unless otherwise specified herein, these tests shall be performed at the daylight viewing area (head Wedge 8289187).

4.6.4.1 Collimation, Pre-Vibration/Pre-Shock, daylight viewing area.- Prior to performing this test and the following tests of 4.6.4.2 and 4.6.4.3, disengage the index pin "D" from the index plate "E" of the test fixture and place the 10 second level on surface "C" of the leveling base. The bubble of the level shall be centered within its graduation lines. If it is not, then adjustment is required in accordance with the procedures specified in paragraphs 7 and 8 of the set-up instructions contained on Drawing F8247391. With the index pin "D" disengaged from the index plate and surface "C" of the leveling base level, view through Support Assembly Collimator 10548767 positioned at the daylight viewing area and observe the projected target image. The intersecting point of the horizontal and vertical reticle lines of the collimator shall be superimposed on the intersecting point of the horizontal and vertical target image lines within the tolerance specified in 3.7.1 as measured on the collimator's micrometer knobs. Record this measurement for use in the subsequent test of 4.6.8.

4.6.4.2 Collimation, Pre-Vibration/Pre-Shock, passive viewing area.- With the index pin "D" disengaged from the index plate "E" and surface "C" of the leveling base levels; view through the auto-collimator positioned at the passive viewing area (Head Wedge 10514254) and observe the projected target image. The intersecting point of the horizontal and vertical reticle lines of the collimator shall be superimposed on the intersecting point of the horizontal and vertical target image lines within the tolerance specified in 3.7.2 as measured on the collimator's micrometer knobs. Record this collimation measurement for use in the subsequent test of 4.6.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.6.4.3 Collimation, unity power area.- Perform this test with the aid of the 26° Elevation Calibration Standard that conforms to Drawing D10548766. The calibration standard shall be set-up in accordance with the set-up instructions specified on Drawing F10547987 and then positioned at the unity power area (Head Window 10514256). With the index pin "D" disengaged from the index plate "E" and surface "C" of the leveling base level, view through the eyepiece of the 3X Telescope 5549108, contained in the calibration standard, and observe the boresight point (intersecting point of the horizontal and vertical reticle lines) of the telescope with respect to the boresight point of the target image. The boresight point of the telescope reticle shall meet the collimation requirements specified in 3.7.3, as measured with reference to the target image.

4.6.4.4 Backlash.- Test test shall be performed subsequent to the test in 4.6.4.3. With the index pin "D" disengaged from the index plate "E", place the 10 second level on surface "C" of the leveling base. With backlash removed for direction of travel, rotate the leveling base by adjusting the leveling rod (Part #62 depicted on Drawing F8247391) until the bubble of the precision level is centered within its graduation lines without overtravel. View through the auto-collimator positioned at the daylight viewing area and position the horizontal and vertical reticle lines of the collimator coincident with the horizontal and vertical target image lines by means of the collimator's micrometer knobs. Continue rotation of the leveling base approximately 2° in the same direction of travel, then reverse rotation of the leveling base (backlash included) until the bubble of the precision level is re-centered within its graduation lines without overtravel. View through the collimator and observe for coincidence of the collimator's horizontal reticle line with respect to the horizontal target image line. Any adjustment of the collimator's micrometer knobs to re-obtain coincidence of the collimator's horizontal reticle line with respect to the horizontal target image line is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.7.7. Upon satisfactory completion of this test, disengage the leveling rod from the slot of the leveling base, engage the index pin "D" in the zero degree position on the index plate "E" and perform the following tests of 4.6.4.5 through 4.6.4.8 inclusive.

NOTE: Should overtravel occur during the test for backlash, the test shall be nullified and the procedure specified for backlash shall again be performed.

4.6.4.5 Exursion range.- With the input coupling positioned as specified in 3.6b and the index pin "D" engaged in the zero degree position on the index plate "E", view through the auto-collimator positioned at the daylight viewing area and, if necessary, position the horizontal and vertical reticle lines of the collimator coincident with the horizontal and vertical target image lines by means of the collimator's micrometer knobs. When coincidence has been established, view through the collimator to the reflector "H on the test fixture. Observe the collimator reticle and note the position of the boresight point (intersecting point of the horizontal and

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

vertical reticle lines) with respect to its reflected image. Disengage the index pin from the zero degree elevation position on the index plate. While observing the boresight point of the real and reflected image, rotate the input coupling and engage the index pin, first in the 22 degree elevation position and then in the 18 degree depression position on the index plate. While the index pin is engaged at each position (22°EL and 18°DP), the noted position of the boresight point with respect to its reflected image shall not deviate in depression or elevation respectively to determine conformance with 3.7.4 (see 4.6.4.7 and 4.6.4.8 when performing this test).

4.6.4.6 Coupling travel.- Rotate the coupling to its maximum elevation, then to its maximum depression stop position. While the coupling is at each stop, back off the coupling for at least 1/4 rotation and then again return the coupling to the stop. Perform this minimum of 3 times. During these rotations, the coupling shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metal click can be heard to verify that coupling rotation is being limited only by means of contact with the coupling stop pin and not by contact of other components (optical or mechanical) to be in compliance with 3.7.4.1.

4.6.4.7 Plumb travel.- This test may be performed at the same time that the test of 4.6.4.5 is performed. While observing the boresight point (intersecting point of the horizontal and verticle reticle lines) of the real and reflected image, rotate the input coupling so that the line of sight is deflected from 22° above to 18° below the noted position of the boresight point established in 4.6.4.5. The boresight point of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.5 for the positions indicated as measured on the collimator reticle (see 4.6.4.8 when performing this test).

4.6.4.8 Elevation accuracy.- Perform this test in conjunction with test in 4.6.4.7 (plumb travel). The horizontal reticle lines of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.6 for the positions indicated.

4.6.4.9 Image tilt.- This test shall be performed with the index pin "D" engaged in the zero degree position on the index plate "E". View through the auto-collimator positioned at the Head Wedge 8289187 and position one extremity of the collimator's vertical reticle line coincident to the vertical target image line. The image of the vertical target line shall be parallel to the collimator's vertical reticle line within the tolerance specified in 3.7.8, as measured on the target. Repeat the above procedure with the auto-collimator positioned at the Head Wedge 10514254. The image of the vertical target line shall be parallel to the collimator's vertical reticle line within the tolerance specified in 3.7.8, as measured on the target.

4.6.5 Resolution, daylight and infrared viewing areas.- The test equipment specified in 4.6.4 is not required for this test (4.6.5) and the test in 4.6.6. Position the head assembly on its side with side "L"

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

of the input coupling positioned at 22° 30 minutes with respect to the four mounting pads. This test shall be conducted utilizing an observation telescope of at least 20 power with an entrance pupil as specified in 3.8.1, a resolving power chart representing the angular subtense for the seconds of arc specified and a supporting medium capable of receiving the observation telescope as well as the head assembly to be tested. Construction and positioning of the resolving power chart shall comply with the "Resolution test" specified in MIL-I-13830 shall contain four line sets as shown in Figure I therein. Position the observation telescope at the Head Wedge 8289187. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Resolution shall meet the requirement specified in 3.8.1. Repeat the above procedure with the observation telescope positioned at the Head Wedge 10514254. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Resolution shall meet the requirement specified in 3.8.1.

4.6.6 Resolution, unity power system.- Perform this test subsequent to 4.6.5, utilizing an observation telescope of at least 3 power with an entrance pupil as specified in 3.8.2 and a resolving power chart representing the angular subtense for the seconds of arc specified. Construction and positioning of the resolving power chart shall be as specified in 4.6.5. Resolution shall be within the tolerance specified in 3.8.2, as determined by viewing through the observation telescope when positioned at the Head Window 10514256.

4.6.7 Vibration/Shock.

4.6.7.1 Vibration "A".- This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the head assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.7.1 and 3.7.2. Subsequent to the test of 4.6.4.1, the head assembly shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and frequency range shall be as specified in 3.9.1. Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the head assembly along three mutually perpendicular axes while the head assembly is maintained in it's normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F11727486). The head assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Mount Periscope:M118E1. The head assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.9.1. During this

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the tests in 4.6.8 and 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement, friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.3, 4.6.4.4 through 4.6.6 exclusive and 4.6.10 through 4.6.12 inclusive.

4.6.7.2 Vibration "B".- This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the head assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.7.1 and 3.7.2. Subsequent to the test of 4.6.4.1, the head assembly shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and frequency range shall be as specified in 3.9.2. Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the head assembly along three mutually perpendicular axes while the head assembly is maintained in it's normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F11727486). The head assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. The head assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.9.2. During this test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the tests in 4.6.8 and 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.9.2, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.3, 4.6.4.4 through 4.6.6 inclusive and 4.6.10 through 4.6.12 inclusive.

4.6.7.3 Shock.- This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope this shock test will not be required when the complete periscope is subjected to shock. Prior to performing this test, the head assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.7.1 and 3.7.2. The shock equipment and test procedure shall be in accordance with Procedure I, of Method 516.2 specified in MIL-STD-810, except conditions such as "G" values, time duration and wave form shock pulse shall be as specified in 3.9.3. The head assembly shall be clamped on the shock equipment by means of an appropriate adapter having the same interface features as Mount Periscope:M118E1. The head assembly shall then be shocked in accordance with the test procedure specified above, except the application of shock along the vertical axis shall be in one direction only (normal operating position of the head assembly). During this test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. Immediately following the shock test and prior to the examination for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the tests in 4.6.8 and 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.9.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement, friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.3, 4.6.4.4 through 4.6.6 inclusive and 4.6.10 through 4.6.12 inclusive.

4.6.8 Collimation, Post-Vibration/Post-Shock, daylight viewing area.- Perform this test utilizing the procedure in 4.6.4.1. Record the collimation measurement. Any difference between the recorded measurement obtained in this test and the recorded measurement obtained in 4.6.4.1 shall not exceed the tolerance specified in 3.10.

4.6.9 Collimation, Post-Vibration/Post-Shock, passive viewing area.- Perform this test utilizing the procedure in 4.6.4.2. Record the collimation measurement. Any difference between the recorded measurement obtained in this test and the recorded measurement obtained in 4.6.4.2 shall not exceed the tolerance specified in 3.11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

4.6.10 Sealing.- This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 and 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the head assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the head assembly by means of the dew point tester. Interior atmosphere of the head assembly shall conform to the requirements of 3.12. Upon satisfactory completion of this portion of the test, internal pressurizing of the head assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the head assembly with dry nitrogen gas, having a dew point at least as low as minus 25°F, to 5.0 $\pm$ 0.10 psig in accordance with the procedure for internal pressure testing on Drawing F8565556. Retain the applied pressure in the head assembly for a minimum of 2 hours. During the first hour, the input coupling shall be rotated through one full excursion of its articulation range. At the end of the 2 hour period there shall be no evidence of leakage in excess of 0.2 psi as indicated on the pressure gage of the testing equipment, to determine compliance with 3.12. Upon satisfactory completion of the sealing test, the head assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the head assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as minus 25°F. When this atmosphere is attained within the head assembly, the head assembly shall then be sealed at the pressure specified in 3.12.

4.6.11 Cleanliness.- The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through each Head Wedge (8289187 and 10514254), inspection for dirt shall be made by viewing an illuminated field provided by 2 each 15 watt desk flourescent lamps held close to the Head Window 10514255. The head assembly shall conform to the requirements of 3.13a and 3.13b.

4.6.12 Interchangeability.

4.6.12.1 Head Assembly with modules (Elbow Assembly, Passive and Body Assembly, Daylight). The cavity at the Head Wedge 8289187 shall be inspected utilizing the maximum and minimum interchangeability gages conforming to Drawings D11732107 and D11732108 respectively, to determine compliance with 3.2a. Upon successful completion of this portion of the test, the cavity at the Head Wedge 10514254 shall be inspected utilizing the minimum and maximum interchangeability gages conforming to Drawings D11729723 and D11732101 respectively, to determine overall compliance with 3.2a.

4.6.12.2 Head Assembly with mount.- The interchangeability test of the head assembly with its mount shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings F11732155 and F11732156. The head assembly shall conform to the requirements of 3.2b.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11727486. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Head Assembly 11727486 is a modular unit of the Tank Gunner's Periscope: M32E1 and M35E1, primarily used in the fire control system of the 105MM Gun Tank M60 Series.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data (see 5.1).
d. Applicable stock number.
e. Provisions for first article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A667

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:33Z
Check the source to verify that this is the current version before use.